



NeuroCogniAI

**ARTIFICIAL INTELLIGENCE
IN
DEEP LEARNING / GENERATIVE /
AGENTIC AI MODELS**

Comprehensive Overview of Deep Radiomics in Neuroimaging For AD Diagnosis

Modality	Biomarker/Feature	Purpose/Application in AD	Deep Radiomics Benefit
Structural MRI (sMRI)	Brain atrophy	Measures hippocampal and medial temporal lobe atrophy	Detects minute volumetric changes using voxel-based analysis.
Functional MRI (fMRI)	Neuronal connectivity	Assesses activity and connectivity changes in AD	Identifies microstructural abnormalities via texture analysis
Diffusion Tensor Imaging (DTI)	White matter integrity	Evaluates microstructural changes in neuronal pathways	Combines MRI with PET for enhanced diagnostic accuracy
Amyloid PET	Amyloid-beta	Identifies amyloid plaque accumulation	Enhances early diagnosis by detecting AD before clinical symptoms appear
Tau PET	Tau protein	Correlates with neurodegeneration and cognitive decline	Assists in predicting MCI to AD conversion
FDG-PET	Glucose metabolism	Detects hypometabolism in AD-prone regions	Identifies subtle imaging patterns via automated feature extraction

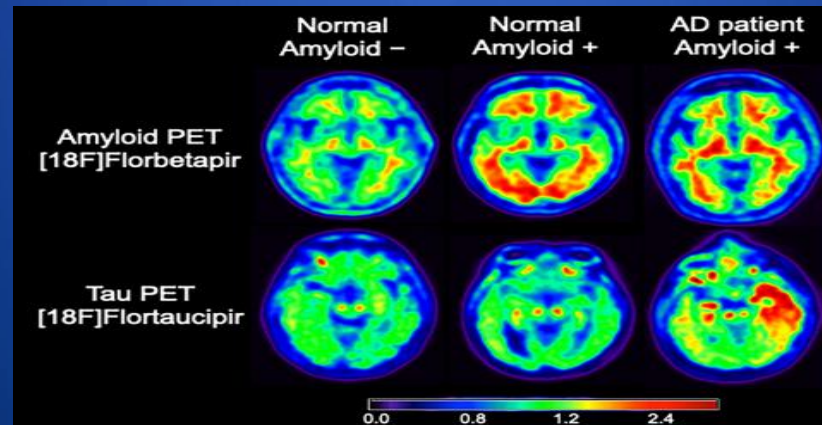
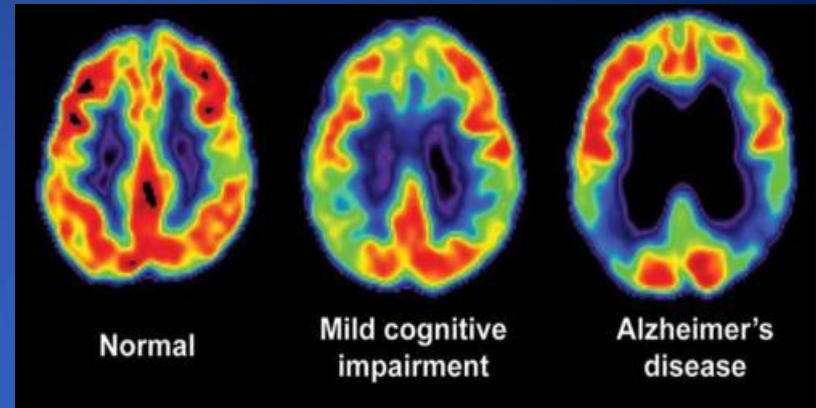
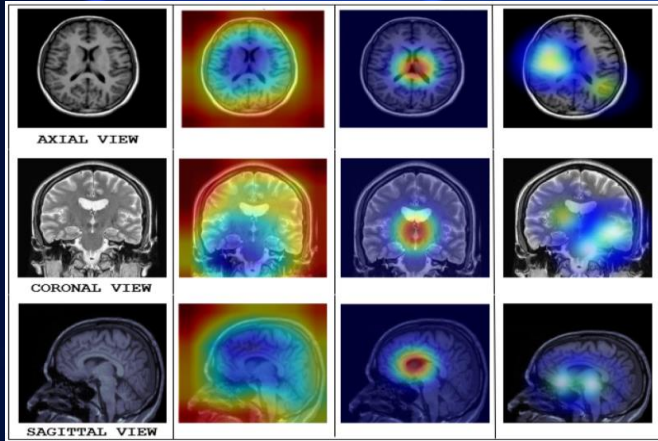
Comprehensive Overview of Neuroimaging Formats For Data Visualisation

Modality	Common Formats	Purpose
Structural MRI (sMRI)	NIfTI (.nii, .nii.gz), DICOM	Anatomical reference (T1/T2 Weighted)
Functional MRI (fMRI)	NIfTI, CIFTI (surface+volume), BIDS	Time-series BOLD activity
Diffusion Tensor Imaging (DTI)	NIfTI, .bval / .bvec (gradient directions), TrackVis (.trk)	White matter tractography
PET Amyloid	NIfTI, DICOM	Beta-amyloid plaque density (e.g., 18F-Florbetapir)
PET Tau	NIfTI, DICOM	Neurofibrillary tau tangles (e.g., 18F-Flortaucipir)

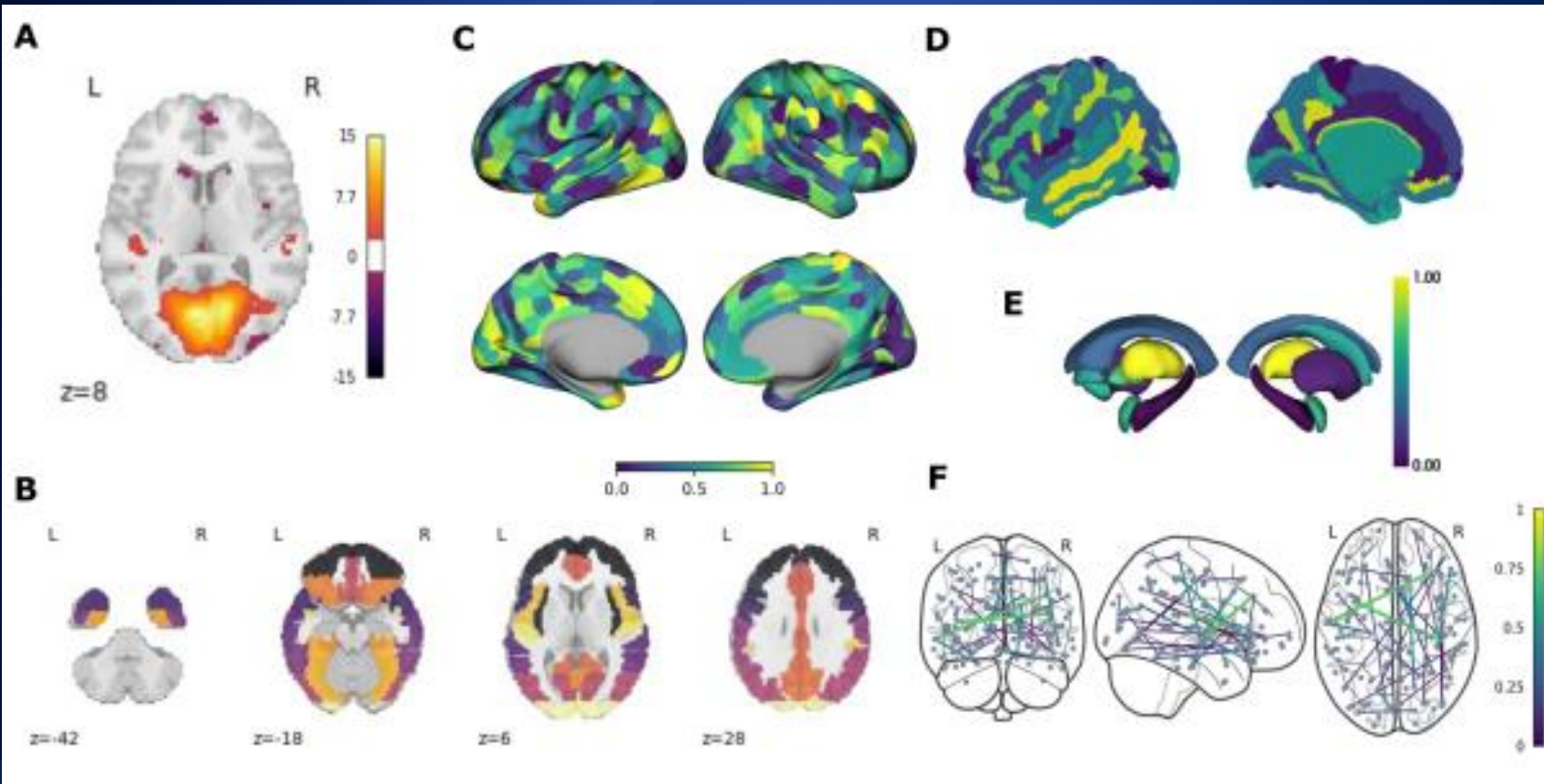
Key format notes:

- **NIfTI** = standard 3D/4D volume with header (voxel size, orientation)
- **DICOM** = raw scanner output, converted to NIfTI for analysis
- **CIFTI** = for cortical surface + subcortical volume (fMRI)
- **BIDS** = directory structure + JSON sidecars for metadata

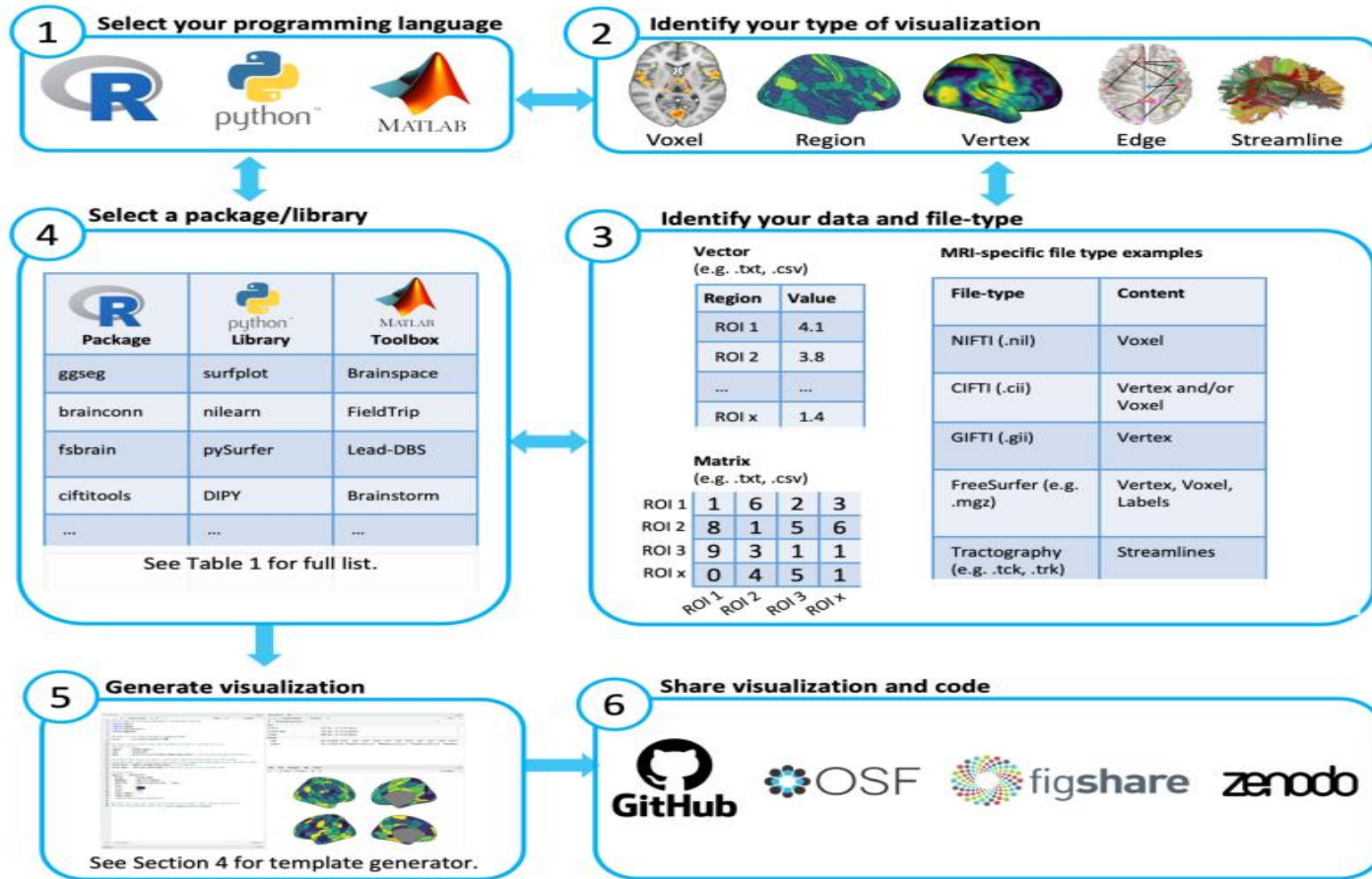
Comprehensive Overview of Neuroimaging Modalities For Data Visualisation



Comprehensive Overview of Neuroimaging Modalities For Data Visualisation



Comprehensive Overview of Neuroimaging Modalities For Data Visualisation



RESEARCH PROSPECTIVES IN NEUROIMAGING [MODELS / XAI / AGENTS]

- ❖ **Build Machine / Deep Learning Models / Frameworks for the classification / interpretation of Neuroimaging Disease**
- ❖ **Comparison of Multi Cohorts Datasets for the classification / interpretation of Neuroimaging Disease**
- ❖ **Build Single / Multi Models For taking Single / Multi modalities**
- ❖ **Simple / Systematic Review of Particular Neuroimaging Models / Frameworks / Disease.**

RESEARCH QUESTIONS & MOTIVATION FOR ALZHEIMER'S DISEASE DIAGNOSIS

SrNo	Research Questions	Motivation
RQ1	What multimodal data combinations are used in the detection of AD?	To know how complimentary modalities improve prediction and characterization
RQ2	What datasets and preprocessing methods are used for multimodal data in Alzheimer's Disease research?	To understand the what preprocessing methods are essential for ensuring data compatibility, reproducibility, and model performance
RQ3	What methods are used for multimodal data fusion in Alzheimer's Disease detection?	By reviewing the fusion methods enables assessment and their effectiveness and suitability for AD diagnosis
RQ4	What different AI models are available for AD diagnosis that incorporates multimodal data?	To identify the best practices and approaches for effectively learning from complex multimodal representation
RQ5	Which XAI frameworks are used to interpret AI models for Alzheimer's Disease using multimodal data?	To explore how XAI techniques enhance trust, transparency, and model validation in a multimodal setting
RQ6	What are the performance trends of AI models in multimodal Alzheimer's Disease diagnosis across different datasets, fusion strategies, and classifiers?	To know the extent to which multimodal fusion and interpretability improve diagnostic performance over traditional methods
RQ7	What are the limitations, challenges, and future scope of using multimodal data and XAI methods for AD diagnosis?	To provide a direction for future research and help to overcome real-world barriers in adoption and generalization

